

Database Design

//1.users table

```
CREATE TABLE users ( user_id INT PRIMARY KEY AUTO_INCREMENT, username  
VARCHAR(50) NOT NULL UNIQUE, email VARCHAR(100) NOT NULL UNIQUE, password  
VARCHAR(255) NOT NULL, role ENUM('student', 'registrar', 'dean') NOT NULL DEFAULT  
'student', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

//2.students table

```
CREATE TABLE students ( student_id INT PRIMARY KEY AUTO_INCREMENT, user_id INT NOT  
NULL UNIQUE, first_name VARCHAR(50) NOT NULL, middle_name VARCHAR(50), last_name  
VARCHAR(50) NOT NULL, phone VARCHAR(20) UNIQUE, sex ENUM('male', 'female') NOT  
NULL, date_of_birth DATE NOT NULL, national_id VARCHAR(50) UNIQUE, address  
VARCHAR(255), student_status ENUM('active', 'inactive') NOT NULL DEFAULT 'active', created_at  
TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (user_id) REFERENCES users(user_id)
```

```
);
```

//3.prorams table

```
CREATE TABLE programs ( program_id INT PRIMARY KEY AUTO_INCREMENT, program_name  
VARCHAR(100) NOT NULL UNIQUE, description TEXT, is_active BOOLEAN NOT NULL  
DEFAULT TRUE, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

//4.Application table

```
CREATE TABLE applications ( application_id INT PRIMARY KEY AUTO_INCREMENT, student_id  
INT NOT NULL, program_id INT NOT NULL, academic_year VARCHAR(9) NOT NULL, status  
ENUM('draft', 'pending', 'approved', 'rejected') NOT NULL DEFAULT 'draft', submitted_at  
DATETIME NULL, reviewed_at DATETIME NULL, reviewed_by INT NULL, created_at  
TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (student_id) REFERENCES students(student_id),  
FOREIGN KEY (program_id) REFERENCES programs(program_id),  
FOREIGN KEY (reviewed_by) REFERENCES users(user_id)
```

```
);
```

//5.document table

```
CREATE TABLE documents ( document_id INT PRIMARY KEY
AUTO_INCREMENT, application_id INT NOT NULL, document_type
ENUM( 'national_id', 'certificate', 'transcript', 'photo' ) NOT NULL, file_name VARCHAR(255) NOT
NULL, file_path VARCHAR(500) NOT NULL, file_type VARCHAR(100) NOT NULL, file_size INT
NOT NULL, uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (application_id)
REFERENCES applications(application_id)
```

```
);
```

//6 payment table

```
CREATE TABLE payments ( payment_id INT PRIMARY KEY AUTO_INCREMENT, application_id
INT NOT NULL, amount DECIMAL(10,2) NOT NULL, payment_method ENUM( 'telebirr', 'CBE')
NOT NULL, transaction_reference VARCHAR(100) UNIQUE, payment_status
ENUM( 'pending', 'paid', 'failed' ) NOT NULL DEFAULT 'pending', payment_date DATETIME
NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (application_id)
REFERENCES applications(application_id)
```

```
);
```

//comments table

```
CREATE TABLE comments ( comment_id INT PRIMARY KEY AUTO_INCREMENT, sender_id
INT NOT NULL, receiver_id INT NOT NULL, message TEXT NOT NULL, is_read BOOLEAN NOT
NULL DEFAULT FALSE, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (sender_id) REFERENCES users(user_id),
FOREIGN KEY (receiver_id) REFERENCES users(user_id)
```

```
);
```